WORK ZONE SAFETY

A STUDY GUIDE ON ROADSIDE & SITE DELINEATION

INTRODUCTION

Work zones are planned environments where construction, maintenance, and utility work meet active traffic corridors. Cones, barriers, and signs are structured lines of defense that organize traffic flow, separate workers from moving vehicles, and secure valuable equipment. Protecting technicians on-site requires understanding site delineation and respecting the boundaries established on the road.

Work zones are not random setups—they are engineered barriers between moving traffic and human lives.

WHY IT MATTERS

- Vehicle intrusion caused by speeding, distraction, or poor visibility is the primary threat to roadside workers.
- Unpredictable public driving, including erratic merges and aggressive reactions, introduces uncontrolled variables into the workspace.
- Heavy equipment operation inside tight boundaries creates serious struck-by and caught-between hazard zones.

Clear site delineation transforms high-risk roadways into protected work areas.



DELINEATE EARLY
HOLD THE LINE

CRITICAL WORK ZONE HAZARDS

Vehicle Intrusion

- Drivers breach boundaries due to speeding or phones
- Erratic lane changes near tapers and lane shifts
- **Fix:** Sturdy barriers and high-visibility advanced signs

Heavy Equipment Blind Spots

- Technicians caught between machinery and fixed walls
- Backing machinery with obstructed rear visibility
- **Fix:** Designated travel paths and audible alarms

Uneven Surfaces & Debris

- Loose gravel, open trenches, and broken asphalt
- Slips, trips, and falls during material transit
- **Fix:** Stable footings and continuous ground checks

Low Visibility Conditions

- Dusk, nighttime work, or bad weather obscuring signage
- Drivers misinterpreting or missing merge cues entirely
- **Fix:** High-lumen work lights and reflective warning signs

Advance warning signs give drivers the seconds needed to react.

THE 5 PILLARS OF WORK ZONE PROTECTION

1. **Establish Advance Warning Boundaries**
   Deploy "Road Work Ahead" or "Work Zone Begins" signage well ahead of the physical perimeter to give oncoming traffic early notification that lane conditions are changing.

2. **Channel and Organize Traffic Flow**
   Use directional signage ("Merge Right", "Lane Closed") paired with tapered cone lines to direct traffic smoothly away from technicians and prevent last-second bottleneck merges.

3. **Enforce Reduced Speed Limits**
   Post regulatory reduced-speed signage. Lowering transit speeds gives distracted drivers the braking distance required to react to lane shifts and site hazards.

4. **Deploy Physical Separation Barriers** THE CRITICAL LINE
   Never rely on visual signage alone when active traffic passes within striking distance. Use impact-rated barriers, barricades, and buffer vehicles to physically block errant vehicles.

5. **Confirm Clear Zone Termination**
   Install "End Work Zone" signage at the perimeter boundary. This clearly communicates to drivers that normal traffic speed and full lane access can safely resume.

A cone is an indicator; an impact barrier is physical protection.

FIELD WORK ZONE SAFETY CHECKLIST

PRE-WORK PERIMETER & DELINEATION VERIFICATION

- [ ] Advance warning signs positioned with adequate sightline
- [ ] Speed reduction and fine notification signs clearly visible
- [ ] Walking and transit surfaces inspected for slip/trip hazards
- [ ] High-visibility Class 2 or Class 3 apparel worn by all personnel
- [ ] Cones and delineators set at standard taper spacing
- [ ] Physical barrier or buffer vehicle staged between traffic and crew
- [ ] Heavy equipment travel paths separated from pedestrian crew zones
- [ ] Auxiliary lighting and reflective equipment activated for dark/dusk

Site Safety Lead _____ Location / Job Site ID _____ Date _____

DELINEATION COMPLETE = AUTHORIZED TO WORK

PLAN THE ZONE, PROTECT THE TEAM

STRUCTURED BOUNDARIES PREVENT ROADSIDE TRAGEDIES